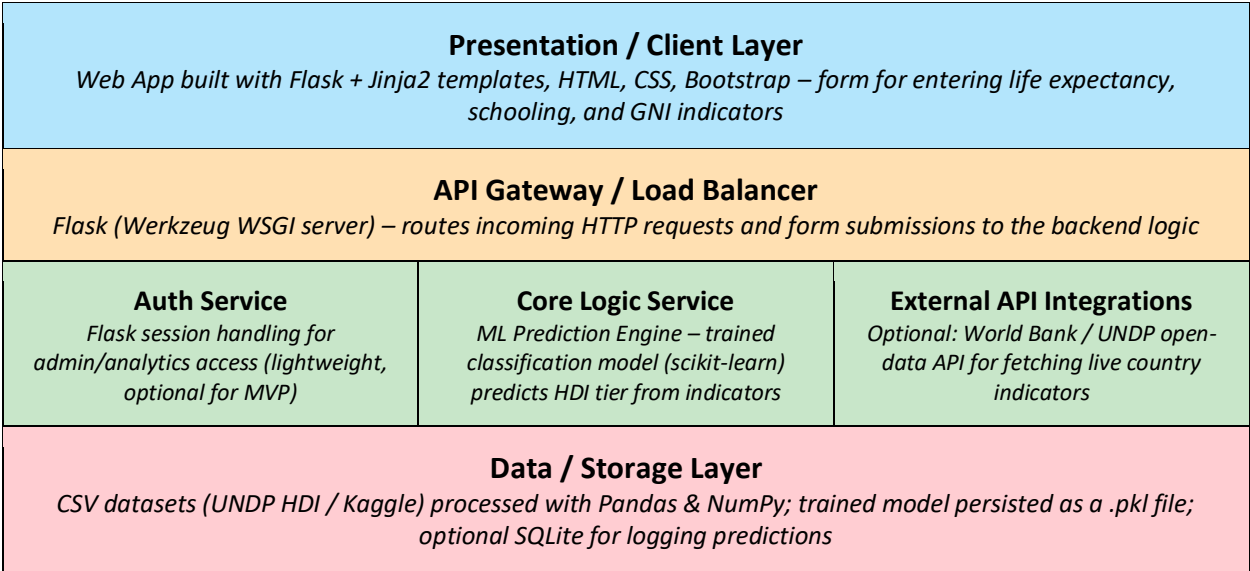


Solution Architecture

Date	28 June 2026
Team ID	
Project Name	A comprehensive measure of well being
Team Members	Chikali Sarika, Anusha Gollapalli, Karishma Sharon Kaparapu, Ambika Chitikela, Parneeth Devanaboina
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture of the HDI Predictor, mapped across the Presentation, Application, and Data layers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects user input (life expectancy, schooling years, GNI per capita) and displays predicted HDI tier and visual insights	HTML, CSS, Bootstrap, Jinja2, Flask
API Gateway	Handles HTTP requests from the client, validates form data, and routes it to the ML core logic service	Flask, Werkzeug WSGI server
Microservices	Loads the trained ML model, preprocesses input indicators, and returns the predicted HDI classification (Very High/High/Medium/Low)	Python, scikit-learn, NumPy, Pandas, Pickle
Database	Stores the historical country indicator dataset used for training	CSV files / Pandas DataFrame, SQLite (optional)

	and, optionally, logs of past predictions	
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